

Pathogenicity of *Beauveria bassiana* against Immature Stages of *Diaphorina citri*¹ under Laboratory Conditions

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Abstract. Asian citrus psyllid, *Diaphorina citri* Kuwayama (Hemiptera: Liviidae), is the most important pest of citrus worldwide. It is the vector of the bacteria *Liberibacter* spp. responsible for Huanglongbing, a devastating disease of citrus. The pathogenicity of four *Beauveria bassiana* (Balsamo) Vuillemin (Hypocreales: Cordycipitaceae) strains and a commercial product were evaluated on immature stages of *D. citri* in laboratory bioassays. At a concentration of 1×10^8 conidia per milliliter, neither the strains nor commercial product of *B. bassiana* had any effect on *D. citri* eggs; 95% of the eggs hatched and nymphs continued development into adults. The strains evaluated, Bb-Rhy, Bb-Dc, Bb-18, Botanigard, and Bb-Hy, killed 58, 56, 45, 43, and 21%, respectively, of all nymphs. The shortest lethal times, 8.5 and 9.5 days, were by Bb-Rhy and Bb-Dc strains. Bb-Rhy killed significantly more I-III than IV-V stage nymphs. Median lethal concentrations of the most pathogenic strains on nymph IV-V instars were 8.34×10^6 conidia per milliliter by Bb-Dc and 1×10^7 by Bb-Rhy. The impact of the fungus on nymphal stages was discussed.

Introduction

Asian citrus psyllid, *Diaphorina citri* Kuwayama (Hemiptera: Liviidae), is the main phytosanitary problem of citrus production worldwide (Hall 2008). It directly damages the plant by feeding on sap from leaves, petioles, and shoots. It is considered economically important because it vectors *Liberibacter* spp. bacteria that cause Huanglongbing (Hall et al. 2012), a devastating disease of citrus (Torres-Pacheco et al. 2013). The vector was first reported in the Caribbean and the United States in 1998. By 2001, it had reached Rio Grande, TX. In Mexico, the insect was initially recorded at Tamaulipas and Nuevo León in the north (Sosa-Armenta et al. 2012) and Campeche and Quintana Roo in the south (Ortega-Arenas et al. 2013). According to Torres-Pacheco et al. (2013), the vector is now in all 23 citrus-producing states of the country, including Chiapas. Huanglongbing was reported first in the states of Yucatán, Jalisco, and Nayarit in 2009 and Sinaloa and Colima in 2010 (López-Collado and López-Arroyo 2013). At Chiapas, Huanglongbing was first reported at Benemérito de las Américas in 2011. By 2015, Huanglongbing had reached the municipalities of Ocosingo, Palenque, Tzimat, La Trinitaria, La Libertad, and Frontera Comalapa (CESAVE-CHIAPAS 2015).

¹Hemiptera: Liviidae

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Alemán et al. (2007) reported systemic and contact insecticides such as imidacloprid, tiametowam, dimethoate, and lambda-cyhalothrin among others were used as chemical control. Biological control with parasitoids (*Tamarixia radiata* Waterston), predators [*Zelus renardii* (Kolenati)] and entomopathogenic fungi [*Hirsutella citriformis* (Speare), *Isaria fumosorosea* Wise, *Metarhizium anisopliae* (Metschnikoff) Sorokin, and *Beauveria bassiana* (Balsamo) Vuillemin] could be alternatives with low environmental impact (Fernández and Miranda 2005). Entomopathogenic fungi are important for biological control and have been used as alternative control against economically important insect pests. Because most insects are susceptible to diseases caused by microorganisms and considering their unique mechanism of invasion, the fungi are excellent control agents as contact insecticides (Téllez-Jurado et al. 2009). *B. bassiana*, a naturally occurring fungal pathogen of many insects in different agroecosystems has been studied extensively as a biological control agent. Several studies reported pathogenicity of *B. bassiana* on Asian citrus psyllid; for example, Ferreira-Pinto et al. (2012) found the median lethal concentration (LC₅₀) of *B. bassiana* on *D. citri* nymphs was 0.4x10⁷ conidia per milliliter. Gandarilla-Pacheco et al. (2013) observed that *B. bassiana* isolates killed 100% of nymph and 80% of adult Asian citrus psyllids. Cruz-Cruz (2013) evaluated pathogenicity of *M. anisopliae*, *H. citriformis*, and three strains of *B. bassiana* against *D. citri* adults and found *B. bassiana* was most pathogenic. Considering the potential of this entomopathogen as a biological control agent, the objective of this study was to evaluate pathogenicity of *B. bassiana* on immature stages of *D. citri* under laboratory conditions.

Materials and Methods

The study was done in biological control laboratories of El Colegio de la Frontera Sur (ECOSUR) at Tapachula, Chiapas, Mexico. *D. citri* was reared and *B. bassiana* strains were reactivated for bioassays.

D. citri adults (females and males) were collected at Colonia La Hermosa Provincia, municipality of Tapachula, Chiapas (N 14°52'48.79", W 92°17'00.22", 120 m above sea level). The insects were transported in 50-ml plastic centrifuge tubes to the laboratory Ecosur Tapachula where they were released in a nursery with 180 *Murraya paniculata* (L.) Jack (Sapindales: Rutaceae) plants covered with organza fabric. One week before releasing adults, the plants were pruned to stimulate shoot growth and thus encourage oviposition and insect development.

B. bassiana strains used were Bb-Dc (isolated from *D. citri* collected at Tapachula, Chiapas), Bb-Hy (isolated from *Hypothenemus hampei* [Ferrari] Coleoptera: Curculionidae collected at a coffee (*Coffea canephora* Pierre) plantation at La Alianza, Cacahoatán, Chiapas), Bb-Rhy (isolated from *Rhyssomatus nigerrimus* Fahraeus Coleoptera: Curculionidae collected at a soybean [*Glycine max* (L.) Merrill] farm at Fracción Tinajas, Tapachula, Chiapas), Bb-18 (isolated from an unidentified lepidopteran larva collected at Papantla, Veracruz, México), and Botanigard, a commercial *B. bassiana* product (Lam International Corporation). Silica gel crystals where different *B. bassiana* strains were preserved were aseptically put into test tubes containing Sabouraud dextrose agar culture medium. The tubes were sealed with Parafilm and incubated at 27 ± 2°C and 75 ± 2% relative humidity for 30 days. The procedure was done for each strain. They were reactivated using 50 adult *Anastrepha ludens* Loew (Diptera: Tephritidae) flies per strain. The flies were submerged for 30 seconds in a suspension of conidia from

each strain, put into 1-liter plastic containers with food (hydrolyzed protein) and water, and incubated in the entomopathogenic fungus reproduction room. Dead flies were removed every 24 hours and put into 9-cm sterilized Petri dishes with a sterile slide and a cotton ball moistened with distilled water and incubated for 10 days. When mycelia appeared on the flies, fungal conidia were extracted using a sterile bacteriological loop and put into test tubes with Sabouraud dextrose agar culture medium. The tubes were sealed with Parafilm and incubated for 30 days.

When conidia were produced, the contents of each test tube were scraped with a bacteriological loop, deposited in a sterile glass vial, and 5 ml of distilled water with 0.05% Tween 80 were added. The suspension in the tubes was shaken in a vortex for 2 minutes every time a dilution was made. For later bioassays, the concentration of each strain was adjusted to 1×10^8 conidia/ml counted in a Neubauer chamber. This concentration was used to determine pathogenicity and LT_{50} , while serial dilutions were made for the LC_{50} .

Sterile slides were used for a viability test. One drop of Sabouraud dextrose agar medium was put at each end where later a drop of conidial suspension was added and covered with a cover slip. The slides were put into Petri dishes with a cotton ball moistened with distilled water. The dishes were sealed with Parafilm and incubated for 24 hours. Three dishes were used for each strain. The proportion of viable conidia was determined by observation at 40X with the aid of a compound microscope. Germinated conidia were those that formed germ tubes. Viability was expressed as percentage of germination (Goettel and Inglis 1997). Viable strains were those in which germination was more than 90%.

Thirty, 8-cm-long branches of *M. paniculata* with turgid, healthy shoots and no longer than 5 cm infested with *D. citri* eggs were selected. Nymphs found were eliminated. Each of the four strains and the *B. bassiana* commercial product were adjusted to a concentration of 1×10^8 conidia/ml and sprayed onto shoots with eggs by using plastic atomizers 3 cm in diameter and 9 cm long. One milliliter of fungal solution was applied per shoot, while the check was treated only with distilled water with 0.05% Tween 80. After applying fungus, the *M. paniculata* branches were put into flat-bottom vials that contained sterile distilled water and stoppered with a cotton ball. The tubes were put into 1-liter plastic containers covered with organza fabric, and incubated. Each treatment was replicated five times. Emergence of nymphs in both treatments and the check was monitored. The effect of the different *B. bassiana* strains on *D. citri* eggs was observed every 72 hours for 13 days. A completely randomized design was used in which experimental units were the *M. paniculata* branches infested with *D. citri* eggs and treatments were the four fungal strains and commercial product and the check.

To determine pathogenicity and LT_{50} of first- through fifth-instar nymphs, methodology described previously was used, but with I-V instar nymphs counted on selected shoots. Each treatment was replicated five times and dead nymphs were counted and put into humidity chambers to stimulate sporulation. The number of nymphs with mycosis (fungal infection) was counted.

The number of *D. citri* fifth-instar nymphs killed by *B. bassiana* strains was determined every 24 hours for 13 days. A completely randomized design was used in which experimental units were *M. paniculata* branches infested by I-V instar nymphs, and treatments were the four strains, commercial product, and check.

Because results from the previous bioassay apparently showed that I-III instar nymphs were scarcely affected by fungal treatments, another bioassay to determine pathogenicity and LT_{50} used one of the most aggressive strains, Bb-Rhy,

and nymphs separated into two groups: I-III instars and another group of IV and V instars. The previous procedure was used.

For bioassays to determine LC₅₀ of strains Bb-Dc and Bb-Rhy, 28 *M. paniculata* branches, 8 cm long with turgid, healthy shoots to 5 cm long infested with IV- and V-instar nymphs (25 nymphs per shoot) were used. Strains Bb-Dc and Bb-Rhy were the most pathogenic and had the shortest lethal time in previous bioassays. The procedure was the same as for previous bioassays, except six concentrations, 1x10⁴, 1x10⁵, 1x10⁶, 1x10⁷, 1x10⁸, and 1x10⁹ conidia per milliliter, were prepared for each strain. The check was distilled water with 0.05% Tween 80. Bb-Dc, Bb-Rhy, and the check were sprayed on the nymphs and incubated at 27 ± 2°C and 75 ± 2% relative humidity. The number of dead *D. citri* nymphs was determined every 24 hours for 13 days. Dead insects were counted and put into humidity chambers to induce fungal sporulation, and the number of nymphs with mycosis was counted. A completely randomized experimental design with treatments of the six concentrations of fungi and a check was used with four replications of *M. paniculata* shoots infested by IV- and V-instar nymphs.

Probit analysis using R software (R Core Team 2016) was used to analyze results of the pathogenicity, LT₅₀, and LC₅₀. Mortality data were corrected using the Schneider-Orelli equation (Zar 1999), analyzed by ANOVA and multiple comparison, and a Tukey test was applied when differences were significant (*P* < 0.05). Survival with interval sensor data was analyzed with a log-rank asymptote and multiple comparison test using the Bonferroni correction. The number of nymphal instars killed was analyzed with a goodness-of-fit test with discrete uniform distribution using the likelihood ratio test (G-test) (Zar 1999).

Results

The four strains and commercial product of *B. bassiana* at a concentration of 1x10⁸ conidia per milliliter had no negative effect on *D. citri* eggs: 95% of eggs hatched and nymphs developed normally. The same was observed with the check. No nonhatched eggs from the treatments or check had mycosis.

The greatest corrected mortality of *D. citri* I-V instar nymphs was caused by a concentration of 1x10⁸ conidia per milliliter of strains Bb-Rhy (58.11%) and Bb-Dc (55.99%) (Table 1). Strains that did not kill 50% of *D. citri* nymphs were Bb-18 (45.61%), Bb-Hy (21.22%), and commercial Botanigard (43.64%). Survival analysis

Table 1. Observed Mortality, Corrected Mortality, and *Diaphorina citri* First- to Fifth-instar Nymphs with Mycosis Caused by Four Strains and a Commercial *Beauveria bassiana* at a Concentration of 1 x 10⁸ Conidia per Milliliter

Treatment	Total nymphs	Observed mortality (% ± SE)	Corrected mortality (%) ^a	Nymphs with mycosis	
				Number	% ± SE
Check	319	11.29 ± 2.53a	-	0	0.00
Bb-18	257	51.75 ± 3.58ab	45.61	4	3.01 ± 0.88
Bb-Dc	228	60.96 ± 3.29a	55.99	81	58.27 ± 2.60
Bb-Rhy	261	62.84 ± 3.49a	58.11	114	69.51 ± 2.64
Bb-Hy	352	30.11 ± 3.85ab	21.22	56	52.83 ± 2.30
Botanigard	274	50.00 ± 3.70ab	43.64	45	32.85 ± 2.46

Means followed by different letters are significantly different at *P* < 0.05.

^aMortality was corrected according to Schneider-Orelli (Zar 1999).

showed significant differences among treatments ($\chi^2 = 211.92$, $df = 5$, $P = 2.2e-16$). Differences between treatments also were found with multiple comparisons (Bonferroni = 0.0033). ANOVA showed Bb-Dc and Bb-Rhy were more pathogenic than the others but not statistically different than each other ($P = 0.3219$), indicating the two strains were the most pathogenic against *D. citri* nymphs. The greatest percentages of mycosis were by Bb-Rhy (69.51%) and Bb-Dc (58.27%) (Table 1). Dead nymphs of the check did not develop mycosis, but those that did not develop mycosis from *B. bassiana* turned red and lost weight; these symptoms were not observed in check nymphs. Sánchez-Peña et al. (2011) observed and reported red color on individuals of diverse species of insects infected by *B. bassiana*, in which case sometimes sporulation does not occur. Median lethal times (95% confidence limits) at a concentration of 1×10^8 conidia per milliliter were 8.5 (8.5-9.5 CI) and 9.5 (9.5-10.5 CI) days for Bb-Rhy and Bb-Dc, respectively. Pathogenic effect of the fungus on nymphs decreased after 8 days. Pathogenicity tests showed the number of nymphs killed differed among the five instars of *D. citri*. Most susceptible to the fungus were instars IV and V ($\chi^2 = 276.80$, $df = 4$, $P = 1 \times 10^{-40}$).

Numbers of nymphs of *D. citri* killed by *B. bassiana* strain Bb-Rhy was different between the two groups of instars. More IV-V instar nymphs (69.4%) were killed than were group I-III instar nymphs with only 18.98% (Table 2).

Statistical analysis and the Bonferroni correction gave significantly different results for the bioassay ($\chi^2 = 128.13$, $P = 2.2e^{-16}$) (Bonferroni = 0.0083). Tukey test showed the proportion of mortality by Bb-Rhy (IV-V) was greatest and statistically different from the other treatments, and that Bb-Rhy (I-III) was similar to check group IV-V but different from check group I-III; the check treatments did not differ from each other (Table 2). More than 50% of nymphs (I-III) treated with Bb-Rhy survived, and thus, it was not possible to calculate LT_{50} . Treatment with Bb-Rhy (IV-V) resulted in less survival with a median lethal time (95% confidence limits) of 5.5 days (3.5-6.5). Percentages of mycosis from Bb-Rhy were 45% (I-III) and 68% (IV-V). Check nymphs did not develop mycosis and continued development into adults; likewise, fungal-treated nymphs that did not die reached adulthood.

The median lethal concentration for strain Bb-Dc, one of the most pathogenic, was 8.34×10^6 conidia per milliliter, with intervals of 1.63×10^6 and 1.41×10^7 . Percentage of mortality increased as the dosage increased, attaining mortality of 91.81% with the greatest concentration (1×10^9) and 12.6% with the least concentration (1×10^4). Nymphs that did not die from effect of the fungus continued development to adults. With the check, 2.3% of the nymphs were dead (Table 3); nymphs that did not die continued development to adulthood. Percentages of

Table 2. Observed Mortality, Corrected Mortality, and *Diaphorina citri* Nymphs Grouped by Instar (I-III, IV-V) with Mycosis Caused by *Beauveria bassiana* Strain Bb-Rhy

Treatment	Total nymphs	Observed mortality (% $\pm$ SE)	Corrected mortality (%) ^a	Nymphs with mycosis	
				Number	% $\pm$ SE
Check (I-III)	117	2.56 $\pm$ 0.76c	-	0	0.00
Check (IV-V)	60	10.00 $\pm$ 1.08bc	-	0	0.00
Bb-Rhy (I-III)	95	21.05 $\pm$ 1.97b	18.98	9	45 $\pm$ 1.29
Bb-Rhy (IV-V)	69	72.46 $\pm$ 3.11a	69.40	34	68 $\pm$ 3.03

Means followed by different letters are significantly different at $P < 0.05$.

^aMortality was corrected according to Schneider-Orelli (Zar 1999).

Table 3. Observed Mortality, Corrected Mortality, and Mycosis of *D. citri* Fourth- and Fifth-instar Nymphs Caused by *B. bassiana* Strains Bb-Dc and Bb-Rhy at Different Concentrations (conidia/ml)

Different concentrations (conidia/ml)					
Treatment	Total	Observed	Corrected	Nymphs with mycosis	
	nymphs	mortality (% $\pm$ SE)	mortality (%) ^a	Number	% $\pm$ SE
<i>Strain Bb-Dc</i>					
Check	87	2.30 $\pm$ 0.70	-	0	0.00
1 x 10 ⁴	89	14.61 $\pm$ 1.67	12.60	3	23.08 $\pm$ 0.76
1 x 10 ⁵	86	30.23 $\pm$ 2.13	28.59	10	38.46 $\pm$ 1.24
1 x 10 ⁶	82	40.24 $\pm$ 2.22	38.83	15	45.45 $\pm$ 1.43
1 x 10 ⁷	83	44.58 $\pm$ 2.26	43.28	20	54.05 $\pm$ 1.52
1 x 10 ⁸	95	60.00 $\pm$ 2.39	59.06	41	71.93 $\pm$ 1.70
1 x 10 ⁹	100	92.00 $\pm$ 1.36	91.81	85	92.39 $\pm$ 1.27
<i>Strain Bb-Rhy</i>					
Check	91	7.69 $\pm$ 1.27	-	0	0.00
1 x 10 ⁴	71	23.94 $\pm$ 1.80	17.60	4	23.52 $\pm$ 0.87
1 x 10 ⁵	71	30.99 $\pm$ 1.95	25.24	7	31.81 $\pm$ 1.09
1 x 10 ⁶	63	41.27 $\pm$ 1.95	36.38	9	34.61 $\pm$ 1.21
1 x 10 ⁷	82	46.34 $\pm$ 2.26	41.87	24	63.15 $\pm$ 1.49
1 x 10 ⁸	64	68.75 $\pm$ 1.85	66.15	32	72.72 $\pm$ 1.48
1 x 10 ⁹	66	74.24 $\pm$ 1.78	72.09	39	79.59 $\pm$ 1.41

^aMortality was corrected according to Schneider-Orelli (Zar 1999).

mycoses were greater at the greatest concentrations. For concentration 1x10⁹, the percentage was 93.39%, and at 1x10⁴, mycosis was 23.08%. Because check nymphs that died did not develop mycosis, it was evident that treatments with *B. bassiana* killed nymphs.

Median lethal concentration for strain Bb-Rhy was 1x10⁷ conidia per milliliter, with intervals of 1.62x10⁶ and 1.81x10⁷. Mortality increased with concentration; greatest (1x10⁹) and least (1x10⁴) concentrations killed 74.24 and 23.94% of nymphs, respectively (Table 3). Nymphs that did not die from fungal infection continued development to adults. Mortality in the check was 7.69%; nymphs that did not die continued development to adulthood. Mycosis was more at the greatest concentration; percentage of mycosis at 1x10⁹ conidia per milliliter was 79.59% and 23.52% at 1x10⁴. Check nymphs that died did not develop mycosis, demonstrating the fungus killed nymphs of the other treatments.

Discussion

This is the first report on lack of impact caused by *B. bassiana* to eggs of *D. citri*. The results showed the four strains and commercial product of *B. bassiana* evaluated had no negative effect on *D. citri* eggs; 95% hatched and continued normal development into nymphal stages. Therefore, *B. bassiana* did not affect the psyllid during egg development. Apparently, eggs of insects are less attacked by entomopathogenic fungus than are other immature stages. Al-Deghairi (2008) found that *B. bassiana* killed few *B. tabaci* eggs (4.49%). The author reported the phenomenon was caused by the structure of the chorion that is a strong barrier against invasion by fungal spores. The fungus germ tube also requires time to germinate and penetrate; in eggs, it is probable the time is longer than that of

embryo development. There also might be antifungal compounds on the egg surface that inhibit germination of conidia (Meeks et al. 2000). Gindin et al. (2000) reported that eggs of *Bemisia argentifolii* Bellows & Perring were immune to infection by *Lecanicillium lecanii* (Zimmermann) Zare & W. Grams (= *Verticillium lecanii*) but when the eggs were ready to hatch, fungal hyphae could infect the next stage. In our study, all nymphs that hatched from eggs treated with *B. bassiana* reached adulthood. However, further evaluation of the effect of other strains of *B. bassiana* and other entomopathogenic fungi is suggested on psyllid eggs before assuming immunity to microorganisms by this stage of development.

In contrast to eggs, *D. citri* nymphs were susceptible to the *B. bassiana* strains tested. Strains Bb-Rhy and Bb-Dcc at a concentration of 1×10^8 conidia per milliliter especially were more pathogenic, killing 58.11 and 55.99%, respectively. Padulla and Alves (2009) obtained 72% mortality of II-IV instars with a strain of *B. bassiana* at a concentration of 5×10^7 conidia per milliliter. Likewise, Torres-Acosta et al. (2011) who evaluated *B. bassiana* with different bioassay techniques and two methods of inoculating I-III instar nymphs, obtained 73% mortality with a concentration of 1×10^8 conidia per milliliter. However, the results show that mortality possibly depends on characteristics of the strains evaluated (different host, different geographical location, etc.). However, we noted that the percentage of nymphs killed was not proportional to the percentage of mycosis; strain Bb-18 killed a greater percentage, but the percentage of mycosis was low. In contrast, the commercial strain that caused a large percentage of mycosis, caused low mortality. The differences might be related to virulence and production of secondary metabolites that intervene in the ability of a pathogen to cause disease (Padulla and Alves 2009), as well as to intrinsic conditions of the pathogen, natural susceptibility or resistance of the insect host, and other factors (Alves 1998).

While mortality caused by pathogenic fungi is important because it assures decrease of pest abundance (Gandarilla-Pacheco et al. 2013, Casique-Valdes et al. 2015, a large percentage of mycosis is equally important because it increases the probability that the pathogen disseminates throughout the field. For that reason, an isolate that causes much mortality and is capable of colonizing and causing mycosis in a large percentage of infected dead insects is a more effective pathogen for control than is one that causes much mortality but whose sporulation on the host is low. In our study, the two strains (Bb-Rhy and Bd-Dc) that were most pathogenic against *D. citri* nymphs also produced large percentages of mycosis and thus were feasible for biological control of the psyllid.

Pathogenicity of the strains was greater for IV and V instars than for instars I-III. Malekan et al. (2012) reported similar results when they compared the effect of *B. bassiana* and *Lecanicillium muscarium* Zare and Gams and in combination with the insecticide imidacloprid on different stages of *Trialeurodes vaporariorum* (Westwood) nymphs. According to the authors, percentage of mortality and amount of infection caused by the fungus were greater for III and IV instars than for instars I and II, indicating the later stages were more susceptible to fungi. Poprawski et al. (2000) also mentioned that III instar nymphs of *T. vaporariorum* were more susceptible than I instars when inoculated with the fungi *Paecilomyces fumosoroseus* (Wize) Brown & Smith and *B. bassiana*. Other studies mention that earliest nymphal instars were more susceptible to certain pathogens, such as *L. muscarium* with *B. tabaci* on tomatoes and verbena. Cuthbertson et al. (2005) observed the II instar of the whitefly was more susceptible to infection caused by fungus. Gindin et al. (2000) indicated high pathogenicity of *L. lecanii* to all

developmental stages except eggs of *B. argentifolii*. Siongers and Coosemans (2003) found commercial Botanigard killed many *T. vaporariorum* first-instar nymphs that we did not observe in our study with *D. citri*. Other reports indicated the developmental phase of the nymphs did not influence the effectiveness of fungal applications. Lacey et al. (2009) assessed insecticidal activity of isolates of *B. bassiana*, *M. anisopliae*, and *I. fumosorosea* on adults and I- and III-stage nymphs of potato psyllid, *Bactericera cockerelli* (Sulc). They concluded the age of nymphs did not affect effectiveness. It is not known why results on susceptibility of nymphs to pathogenic fungi vary. Malekan et al. (2010) suggested for *T. vaporariorum*, differences in mortality were possibly caused by nymph molting in the first stages shortly after inoculation by fungi and also to the permanence of conidia on later nymphal stages.

In pathogenicity tests of five unknown nymphal stages of *D. citri*, median lethal time at a concentration of 1×10^8 conidia per milliliter was 8.5 days for the Bb-Rhy strain. However, in the bioassay in which nymphs were classified by instar, LT_{50} was 5.5 days, which was possible to calculate only for IV- and V-instar nymphs because 50% of I-, II-, and III-instar nymphs did not die. The difference was because IV- and V-instar nymphs were more susceptible to *B. bassiana*, but when all instars were evaluated together, LT_{50} increased. LT_{50} for IV and V instars in this study was similar to that reported by Ferreira-Pinto et al. (2012) who found LT_{50} of 5.7 days with 1×10^9 conidia per milliliter of *B. bassiana* on *D. citri* nymphs. Padulla and Alves (2009) found for II-V instar *D. citri* nymphs, TL_{50} was 4.71 days.

The median lethal concentrations of strains that were pathogenic was 8.34×10^6 Bb-Dc and 1×10^7 Bb-Rhy conidia per milliliter after 6 days. Padulla and Alves (2009) found the lethal concentration of *B. bassiana* on II-IV instar *D. citri* nymphs was 2.3×10^7 conidia per milliliter. However, Ferreira-Pinto et al. (2012) found the LC_{50} of *B. bassiana* on *D. citri* nymphs was 0.4×10^7 conidia per milliliter after 10 days. Concentrations of *B. bassiana* on *D. citri* nymphs in the studies were similar and more efficient than those by Dwiastuti and Kurniawati (2007) for *H. citrifomis* fungus on *D. citri* nymphs, with lethal concentrations of 2.26×10^{13} on III-instar nymphs and 1.31×10^{13} conidia per milliliter for IV-instar nymphs. The authors also could not estimate LC_{50} for V-instar nymphs. Future research should evaluate the effectiveness of Bb-Rhy and Bb-Dc strains under field conditions to determine their potential as biological control agents against *D. citri*.

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